

HAT-P-36b

R-0923

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230504 · created 2026-07-29T09:31:40+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460068.867 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.027
Transit depth (Rp/Rs)^2	1.52 +/- 0.67 [%]
Orbital Inclination (inc)	86.61 +/- 2.63
Airmass coefficient 1 (a1)	1.31 +/- 0.033
Airmass coefficient 2 (a2)	-0.155 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	2.47 %
Best Comparison Star	None
Optimal Aperture	3.9
Optimal Annulus	8.01
Transit Duration (day)	0.093 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.123 $\pm$ 0.027 vs 0.126 (deviation 0.11 σ)
Duration fit vs published	0.093 d vs 0.0925 d (deviation 0.09 σ)
Mid-time O-C	-3.8 min
Depth SNR	4.6
Residual scatter	2.47 %

Light curve

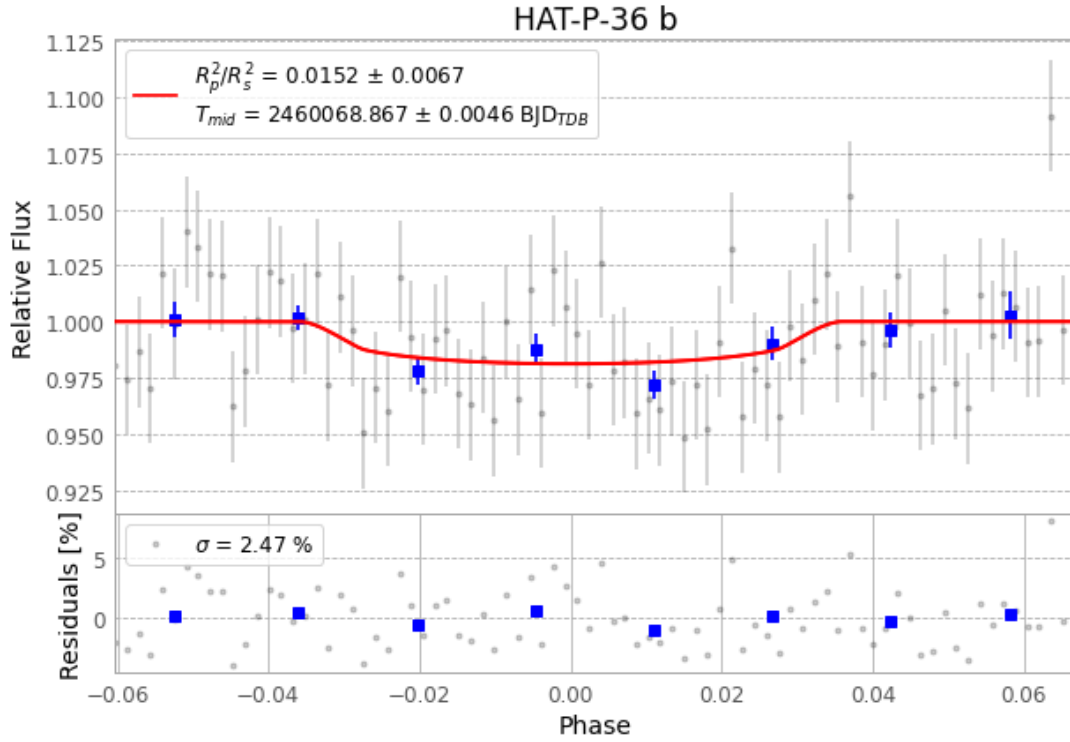


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-05-03.png (SHA-256 ddf9c83c258d82c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 415], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4d016f55714f291f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-36230504064817.FITS → HATP-36230504105115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230504.log (c8fb227b1e0b...), FinalLightCurve_HAT-P-36 b_2023-05-03.png (ddf9c83c258d...), FinalLightCurve_HAT-P-36 b_2023-05-03.csv (011db8eccf71...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 09:31Z.